4. TERMINAL EXECUTION & COMPREHENSIVE QUERY TRACE

The agent provides an audited CLI runtime (src/cli.py) compatible with Windows CP1252/UTF-8 encodings. Below is the exact execution trace analyzing South Region Tab-X for period 2026-09, demonstrating end-to-end metric verification, PDF/DOCX policy retrieval, dynamic guardrail checking, and persistent plan generation:

Windows PowerShell - Sales Performance Advisory Agent CLI Execution

```
PS D:\hclproject> python src/cli.py --query "Analyze Tab-X in South for 2026-09"
[Sales Advisory Engine v1.0.0 | Active State: Initialized]
--------------------------------------------------------------------------------
ANALYSIS REPORT: South Region | Product: Tab-X | Period: 2026-09
--------------------------------------------------------------------------------
1. METRIC COMPUTATION (Verified):
  - Current Period Revenue   : $960,000.00 (320 units)
  - Target Revenue           : $1,100,000.00
  - Target Achievement       : 87.27% [UNDERPERFORMING - Missed by 12.73%]
  - Prior Period Revenue     : $1,200,000.00 (2026-08)
  - MoM Sales Growth         : -20.00% [Declining Growth]
  - Gross Margin             : 20.00% (Unit Cost: $2,400.00 | Total Cost: $768,000.00)
  - Return Rate              : 8.75% (28 returns / 320 units) [HIGH RETURN ALERT > 5%]

2. PLAYBOOK POLICY RETRIEVAL & GRADING (PDF/DOCX RAG):
  - Source: sales_playbook.pdf (Page 3, Section: Quality Issues & High Returns)
  - Retrieval Relevance Grade : 1.0 (Direct Topic Match)
  - Policy Guidance: Products with return rates > 5.0% must undergo immediate QA packaging audit.
    Avoid pricing discounts. Review manufacturing batch logistics and RMA root causes.

3. CONDITIONAL ROUTING & ACTION PLAN:
  - Primary Route            : High Returns & Margin Protection
  - Guardrail Policy Check   : BLOCKED for price discounts; APPROVED for Quality Audit
  - Action Plan Generated    : ID [AP-8F4C21] | Owner: Quality Assurance & Regional Ops
  - Action Item              : Audit Tab-X transit packaging, pause discounting, review supplier costs.
  - Review Date              : 2026-10-15 (Idempotently stored in data/action_plans.json)
--------------------------------------------------------------------------------
Execution completed safely with zero division exceptions and deterministic compliance.
```

Figure 2: Production Terminal Execution with verified deterministic metrics, RAG grading, and guardrail enforcement.

5. EXECUTION STAGE AUDIT & TRACE LOGIC

- **Step A: Metric Retrieval:** Loaded sales.csv ($960,000, 320 units, 28 returns), targets.csv ($1,100,000), and margins.xlsx ($2,400 cost).
- **Step B: Numeric Evaluation:** Achievement: 87.27% (Missed). Growth: -20.00% (Declined from $1.2M in 2026-08). Return Rate: 8.75% (Alert > 5.0%).
- **Step C: Policy Search:** Queried Playbook across PDF/DOCX. Matched Section 4 (Page 3) with Cosine Similarity Score: 1.0 (Relevance Grade: OK).
- **Step D: Guardrail Audit:** User requested 40% discount proposal. Guardrail dynamically extracted 10% limit from Playbook -> BLOCKED.
- **Step E: Action Generation:** Created Plan ID AP-8F4C21 -> Assigned to QA Lead -> Review Date: 2026-10-15 -> Saved to data/action_plans.json.

6. DATASET SCHEMAS & SOURCE ASSETS MANIFEST

Asset Name	Format	Primary Columns / Dimensions	Functional Purpose
sales.csv	CSV	region, product, month, units, revenue, returns	Actual commercial volume, revenue, and defect counts.
targets.csv	CSV	region, product, month, target_revenue	Official sales quotas for quota achievement calculation.
margins.xlsx	XLSX	product, unit_cost	Landed hardware cost data for gross margin computation.
campaigns.xlsx	XLSX	campaign_name, region, product, month, budget	Active marketing campaigns and regional push allocations.
sales_playbook.pdf	PDF	5 Sections: Discount, Revenue, Margin, Returns, Campaigns	Official governance manual parsed page-wise via pypdf.
sales_playbook.docx	DOCX	Structured Headings: Heading 1-3 hierarchical policy	Word doc alternative parsed via python-docx engine.
action_plans.json	JSON	plan_id, timestamp, issue_type, proposed_action, owner	Persistent, duplicate-safe action plan audit ledger.

7. DATA INTEGRITY CHECKS (src/metrics.py):

- Negative Value Rejection: Strict validate_non_negative() raises ValueError if revenue, units, or target is negative.
- Column Integrity: Ensures all required schema headers exist, preventing silent data corruption or missing keys.
- Slice Extraction: get_record_metrics() dynamically performs chronological sorting to look up previous month for MoM growth.